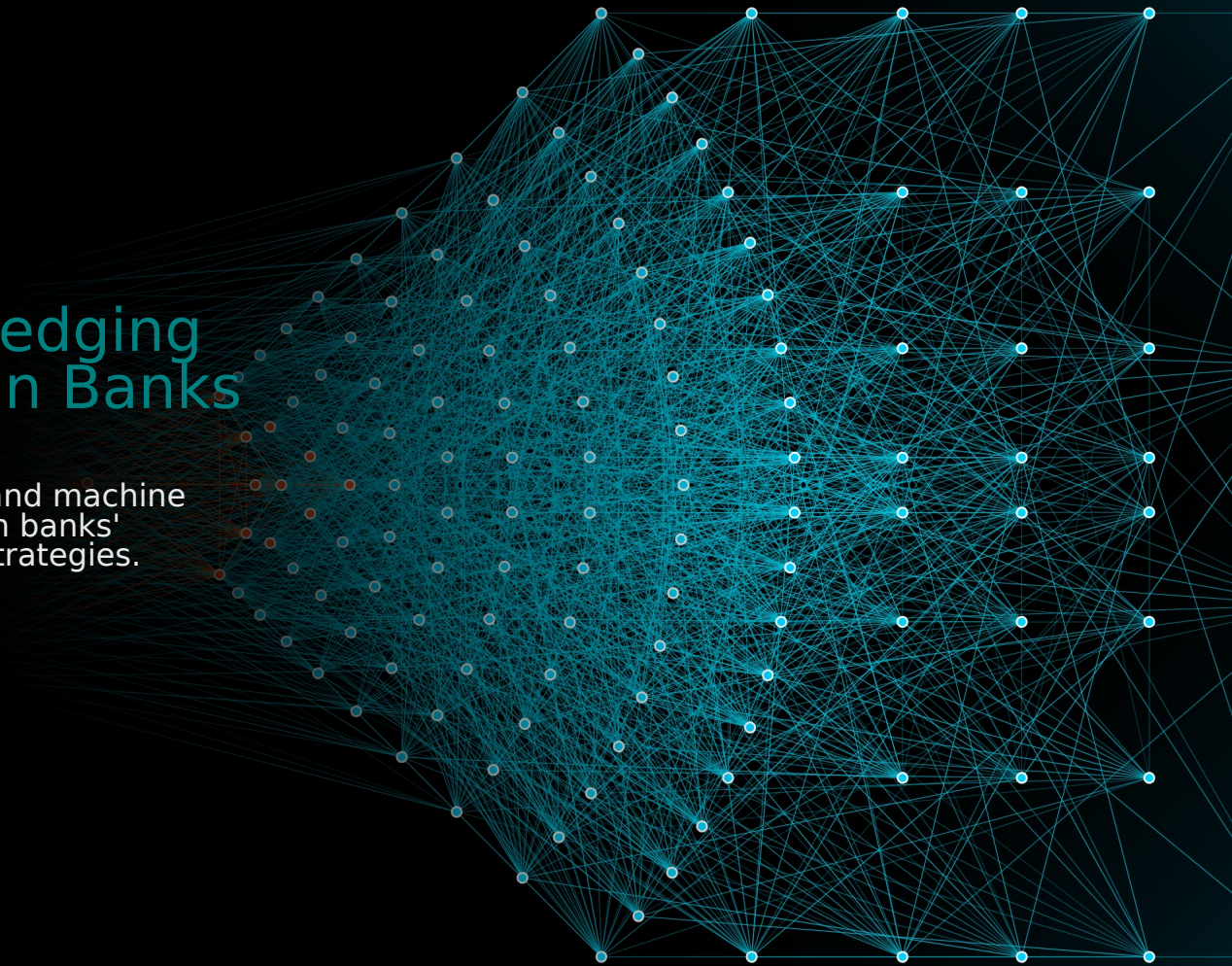


Predicting Swap-Hedging Adoption in Russian Banks

An analysis of financial indicators and machine learning models to forecast Russian banks' decisions to adopt swap-hedging strategies.

Nikkat Afrin



AGENDA

- Introduction and Research Questions
- Dataset
- Exploratory Data Analysis (EDA)
- Data Preparation & Feature Selection
- Machine Learning Models
- Preferred Model & Test Results
- Ensemble Method & Performance
- Challenges & Solutions
- Conclusion



Introduction & Research Questions



Project Motivation

In today's volatile financial landscape, effective risk management is essential for banking stability. Russian banks often adopt swap-hedging strategies to manage interest rate risks. Identifying the key financial indicators influencing this adoption is crucial for risk managers and regulators.



Research Question 1

Which financial performance indicators significantly impact a bank's decision to adopt swap-hedging?



Research Question

Can we ²accurately predict the adoption of swap-hedging in Russian banks using machine learning?



Research Question 3

Does an ensemble method outperform individual machine learning models?

Dataset & Problem Statement



Russian banks (2016-2021)

Dataset covering 2,354 observations of Russian banks over a 6-year period. Financial indicators of Russian banks for 2016-2021



Target: Hedge_indicator (0 = No Hedging, 1 = Hedging)

The binary target variable indicating whether a bank has adopted swap-hedging or not



Financial metrics (Loans, Securities, EBT, Status)

1. Bank status (Active/Inactive) and Year
- 2. Performance Ratios: `ROE`, `ROA`, `H1.0`, and `H4`
3. Logarithmic Variables: `LN\_Loans`,



Objective: Identify financial indicators and accurately predict hedging behavior

The main goal of the project is to understand the drivers of hedging adoption and build a predictive model

The dataset provides a comprehensive view of Russian banks' financial performance and hedging activities, allowing us to explore the key drivers and build predictive models for swap-hedging adoption.

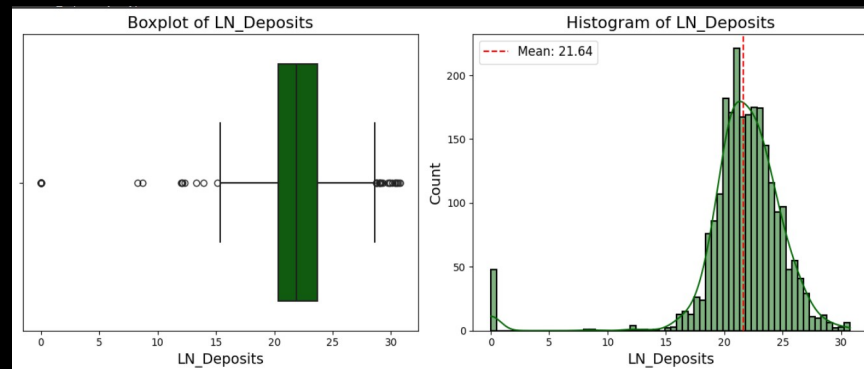
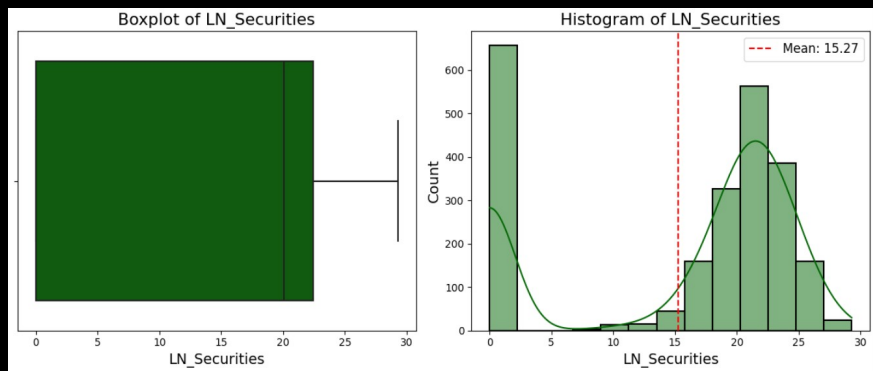
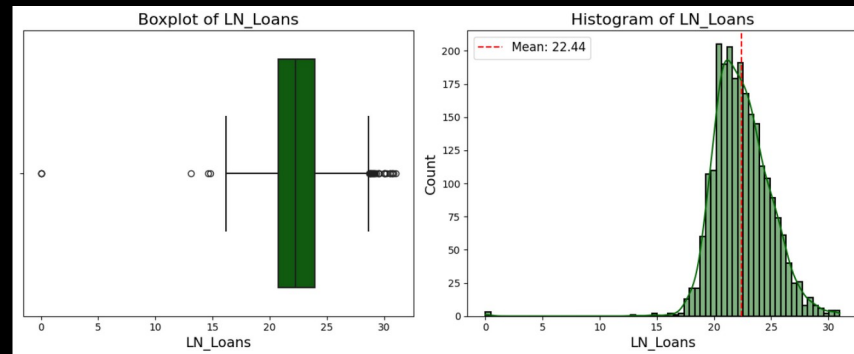
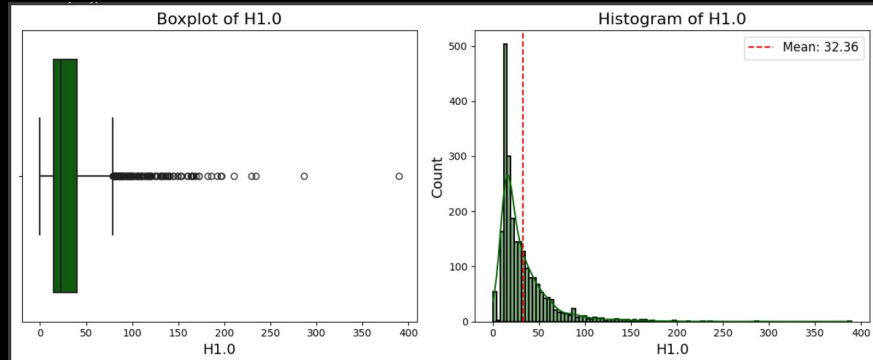
Analyzing the Data - Data Info and Type Casting

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Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype
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0   Reg                    2354 non-null  int64
1   Bank                   2354 non-null  object
2   Status                 2354 non-null  object
3   Year                   2354 non-null  int64
4   H1.0                   2332 non-null  object
5   ROE                    2349 non-null  object
6   LN_Loans               2354 non-null  float64
7   LN_Securities          2354 non-null  float64
8   LN_Deposits            2354 non-null  float64
9   LN_Equity              2347 non-null  float64
10  LN_EBT                 2354 non-null  float64
11  H4                     1993 non-null  object
12  ROA                    2354 non-null  object
13  Overdue_debt_ratio     2332 non-null  object
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memory usage: 276.0+ KB
```

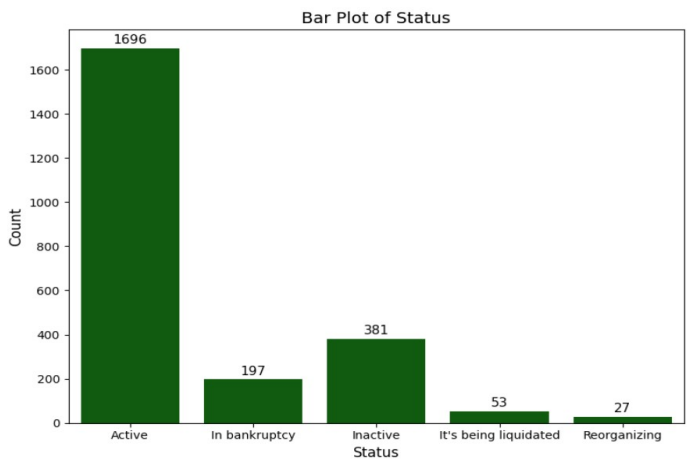
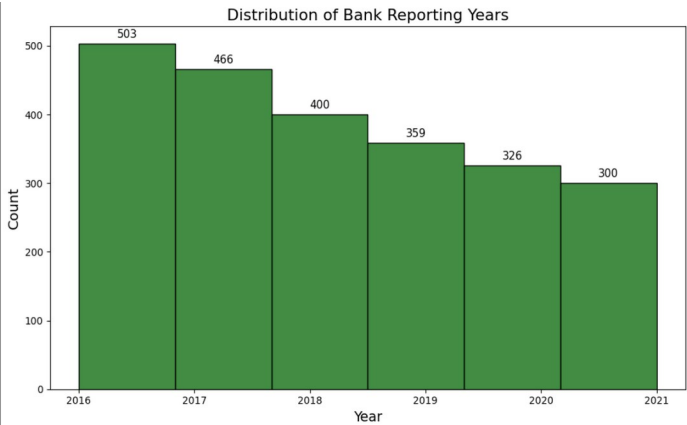
	Missing Count	Missing %
H1.0	22	0.93
ROE	5	0.21
LN_Equity	7	0.30
H4	361	15.34
Overdue_debt_ratio	22	0.93

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2354 entries, 0 to 2353
Data columns (total 15 columns):
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3   Year                   2354 non-null  int64
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5   ROE                    2349 non-null  float64
6   LN_Loans               2354 non-null  float64
7   LN_Securities          2354 non-null  float64
8   LN_Deposits            2354 non-null  float64
9   LN_Equity              2347 non-null  float64
10  LN_EBT                 2354 non-null  float64
11  H4                     1993 non-null  float64
12  ROA                    2354 non-null  float64
13  Overdue_debt_ratio     2332 non-null  float64
14  Hedge_indicator        2354 non-null  int64
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memory usage: 260.1+ KB
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Exploratory Data Analysis (EDA) - Univariate Analysis (Numerical Data)



Exploratory Data Analysis (EDA) - Univariate Analysis (Categorical Data)

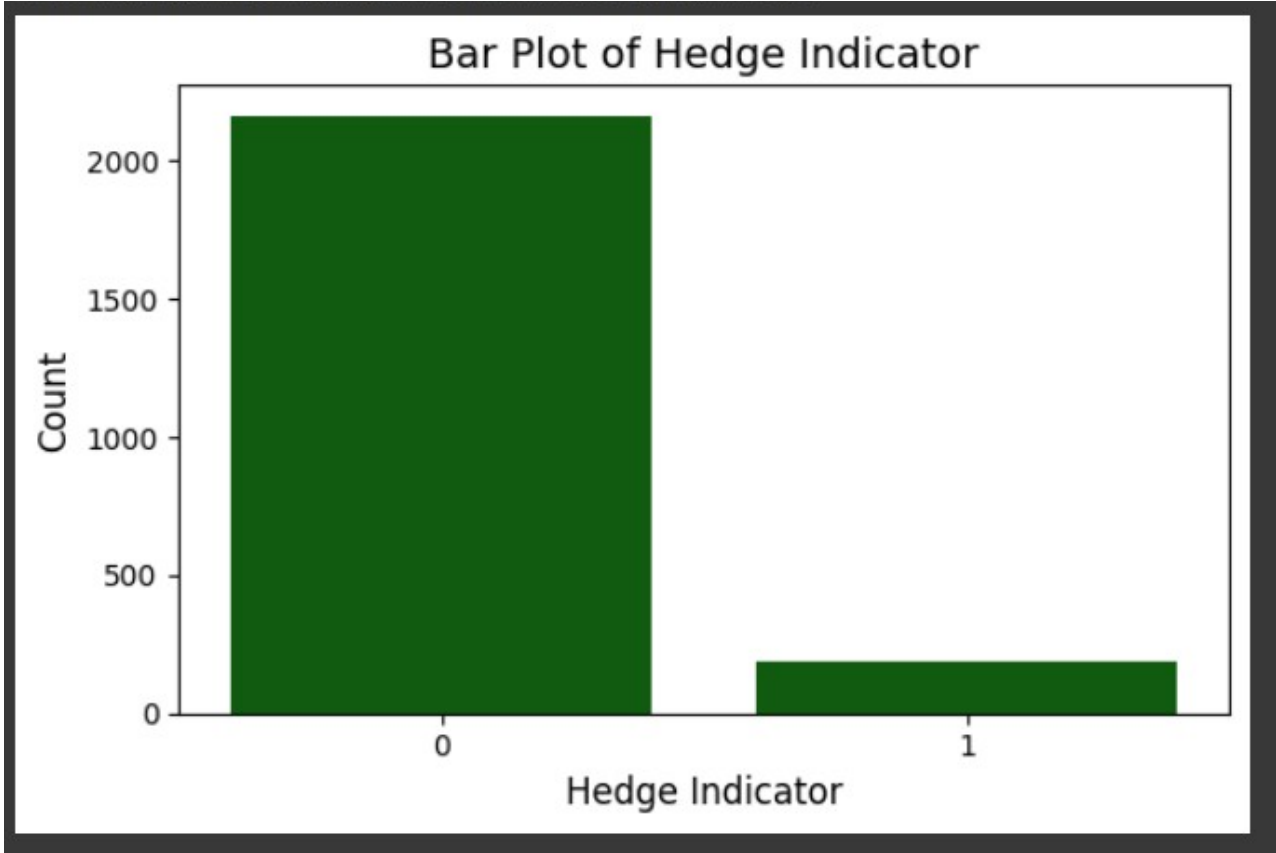


index	Bank	Frequency	Years_Active	Last_Year	Last_Reported_Status
0	AB ROSSIJ, AO	6	2016,2017,2018,2019,2020,2021	2021	Active
1	ABSOLUT BANK (PAO), AKB	6	2016,2017,2018,2019,2020,2021	2021	Active
2	AGORA OOO, BANK	6	2016,2017,2018,2019,2020,2021	2021	Active
3	AGROPROMKREDIT, AO KB	6	2016,2017,2018,2019,2020,2021	2021	Active
4	AGROSOLJZ, OOO KB	2	2016,2017	2017	In bankruptcy
5	AJSIBISI BANK (AO)	6	2016,2017,2018,2019,2020,2021	2021	Active
6	AJVI BANK (AO)	1	2016	2016	In bankruptcy
7	AK BANK, AO	2	2016,2017	2017	In bankruptcy
8	AK BARS BANK, PAO	6	2016,2017,2018,2019,2020,2021	2021	Active
9	AKB KOR, AO	2	2016,2017	2017	Inactive
10	AKSONBANK, OOO KB	2	2016,2017	2017	In bankruptcy
11	AKTIV BANK, AO	5	2016,2017,2018,2019,2020	2020	In bankruptcy
12	AKTSENT, PAO AKB	2	2016,2017	2017	Inactive
13	AL BAALJANS, OOO KB	5	2016,2017,2018,2019,2020	2020	It's being liquidated
14	ALFA-BANK, AO	5	2017,2018,2019,2020,2021	2021	Active
15	ALEF-BANK, AO AKB	6	2016,2017,2018,2019,2020,2021	2021	Active
16	ALEKSANDROVSKI, PAO BANK	6	2016,2017,2018,2019,2020,2021	2021	Active
17	ALMAZERGIBANK AO, AKB	6	2016,2017,2018,2019,2020,2021	2021	Active
18	ALTAJBIZNES-BANK (AO), AKB	2	2016,2017	2017	Inactive
19	ALTAJKAPITALBANK, OOO KB	6	2016,2017,2018,2019,2020,2021	2021	Active
20	ALYNBANK, OOO	6	2016,2017,2018,2019,2020,2021	2021	Active
21	ALZHAN, OOO KB	2	2016,2017	2017	Inactive
22	AMERIKEN EKSPRESS BANK, OOO	6	2016,2017,2018,2019,2020,2021	2021	Active
23	ANELIK RU, OOO KB	1	2016	2016	Inactive
24	ANKOR BANK (AO)	1	2016	2016	In bankruptcy

Show 25 per page

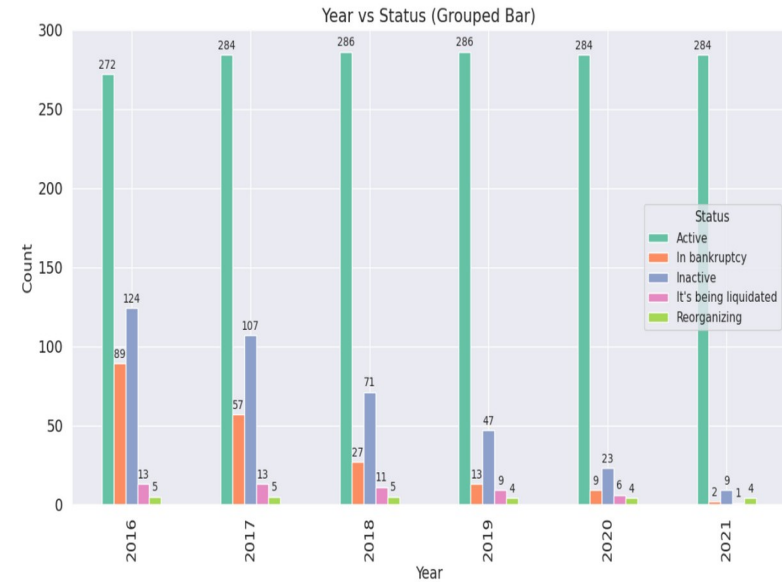
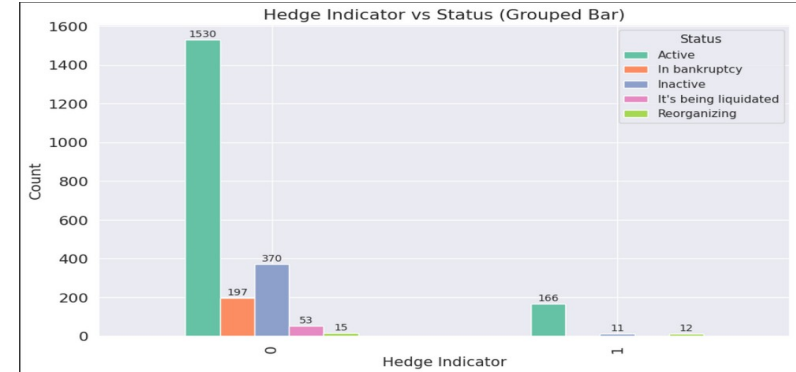
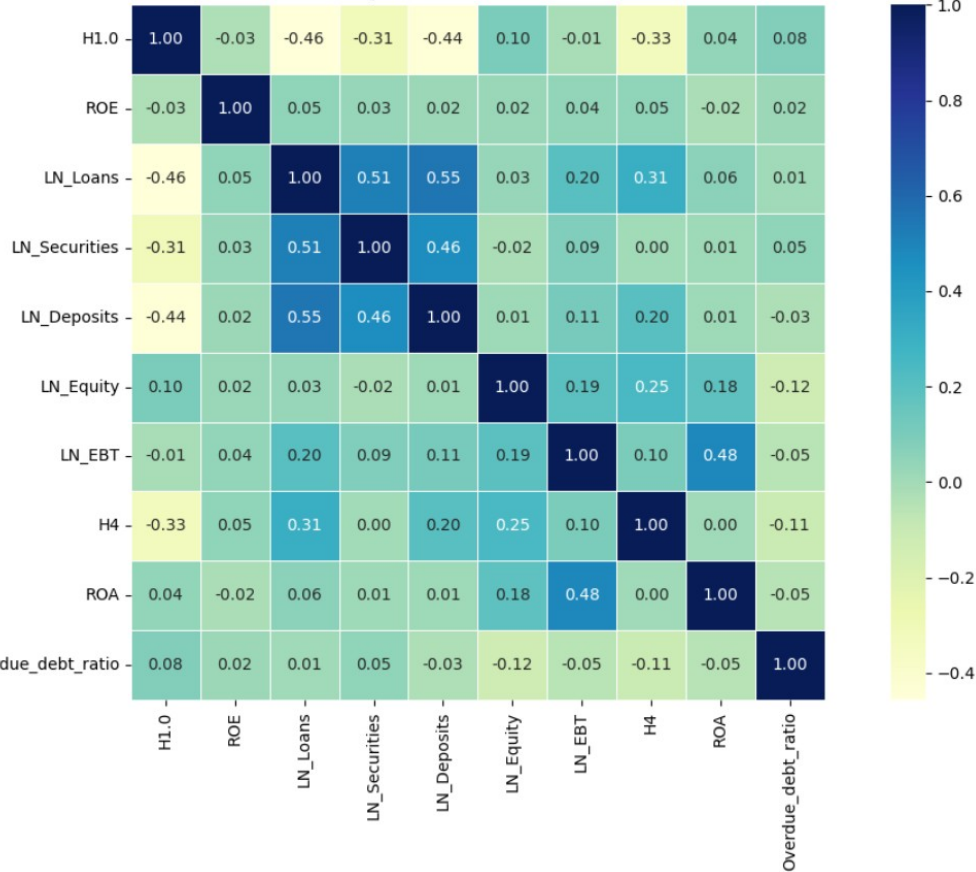
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Exploratory Data Analysis (EDA) - Univariate Analysis (Target Variable)



Exploratory Data Analysis (EDA) - Bivariate Analysis

Correlation Heatmap of Continuous Numerical Variables



Data Preparation & Feature Selection



Data Cleaning

Imputed missing values using median and KNN imputation



Feature Selection

Used Forward Selection to identify the most informative features



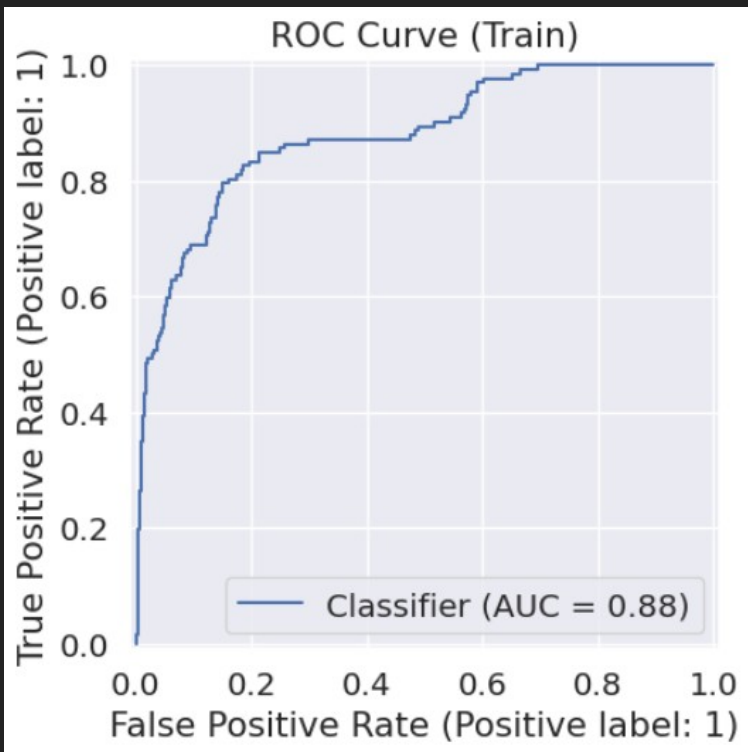
Selected Features

LN_Securities, LN_EBT, LN_Loans,
Status_Reorganizing

Individual Models & Hyperparameters

Model	Hyperparameters
Logistic Regression	Default (Baseline)
Decision Tree	max_depth=3, min_samples_split=2
XGBoost	learning_rate=0.01, max_depth=3, n_estimators=200
Neural Network	SiLU activation, 2 layers, 16 neurons, 200 epochs

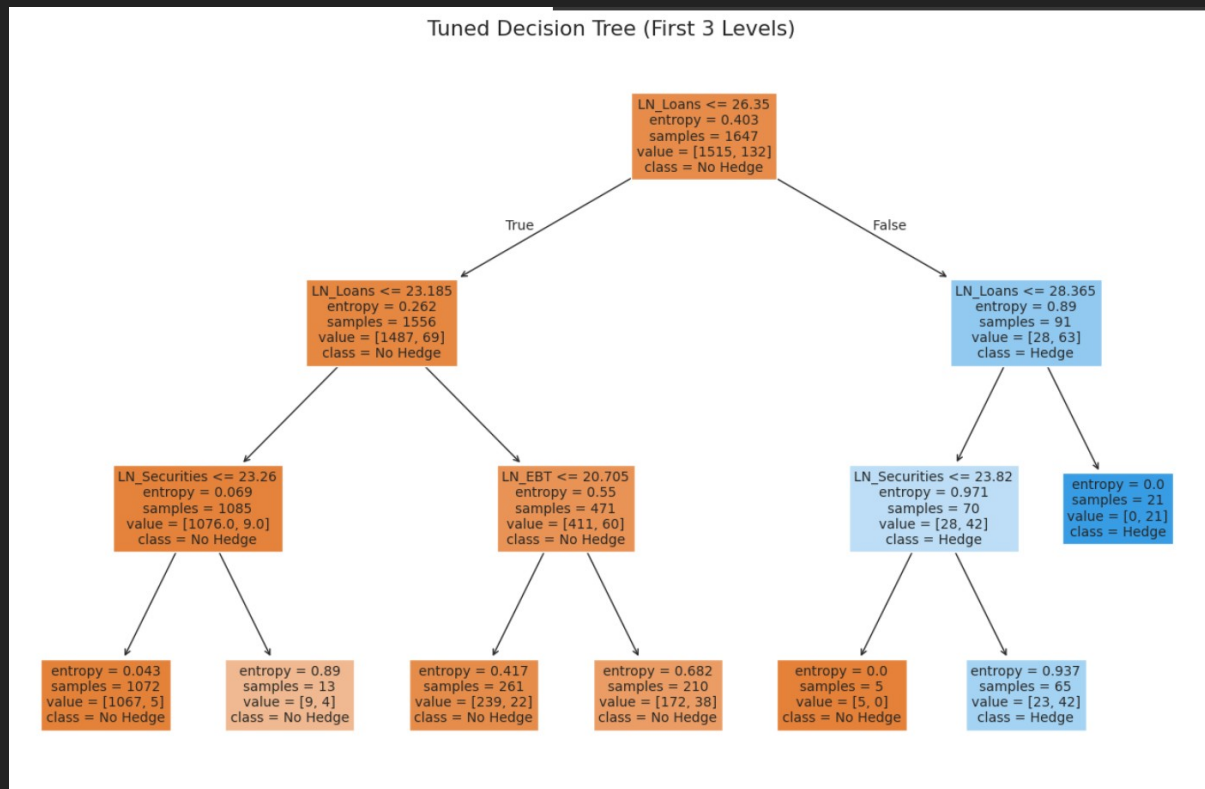
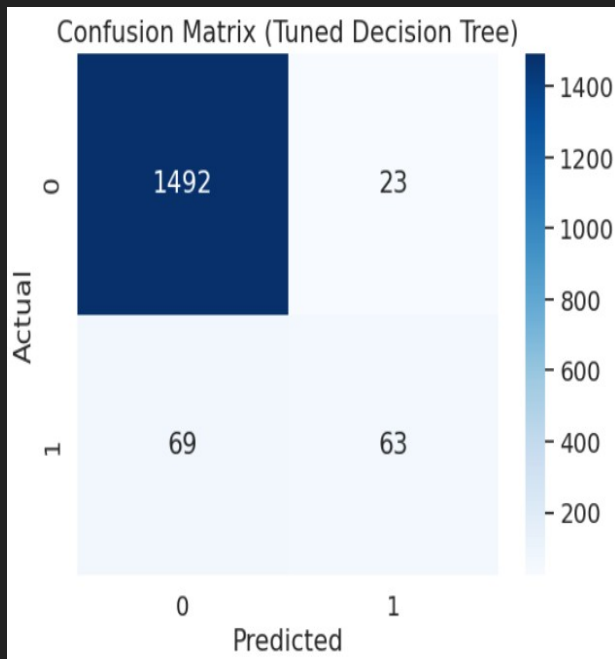
Logistic Regression Classification Report



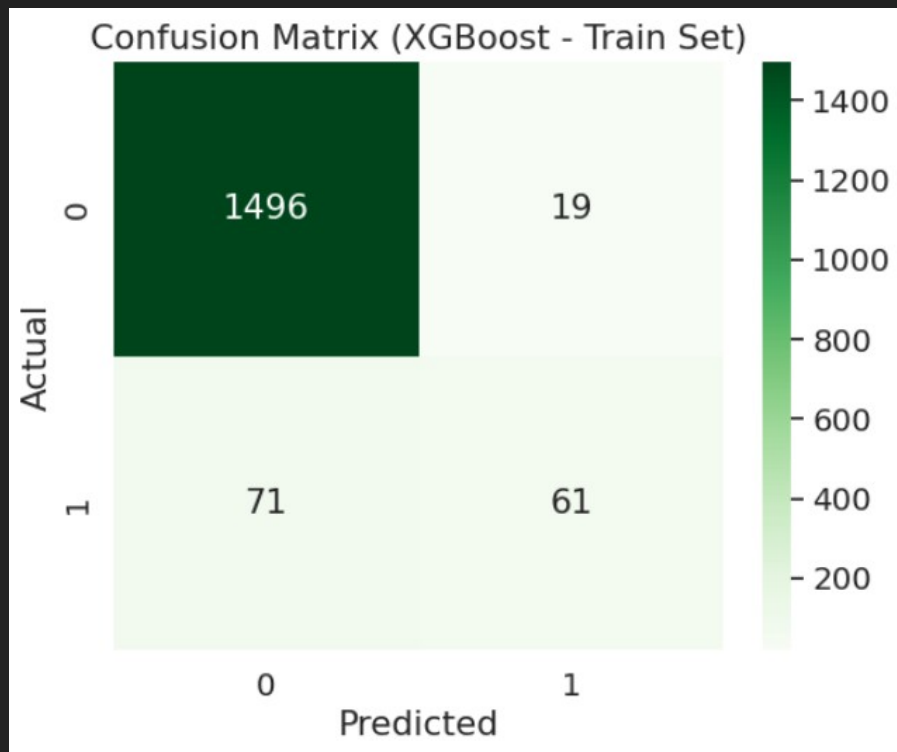
Metric	Value
Accuracy	0.9229
F1 Score	0.1241
AUC	0.8810
Precision (Class 0)	0.92
Precision (Class 1)	0.69
Recall (Class 0)	1.00
Recall (Class 1)	0.07
F1-Score (Class 0)	0.96
F1-Score (Class 1)	0.12
Weighted Avg Precision	0.91

*Logistic Regression Training Metrics

Decision Tree Model



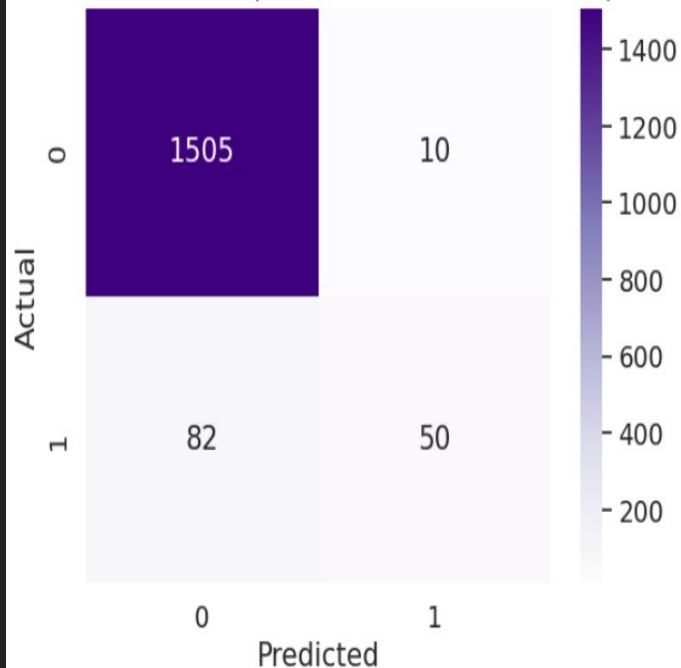
XG Boost Classification



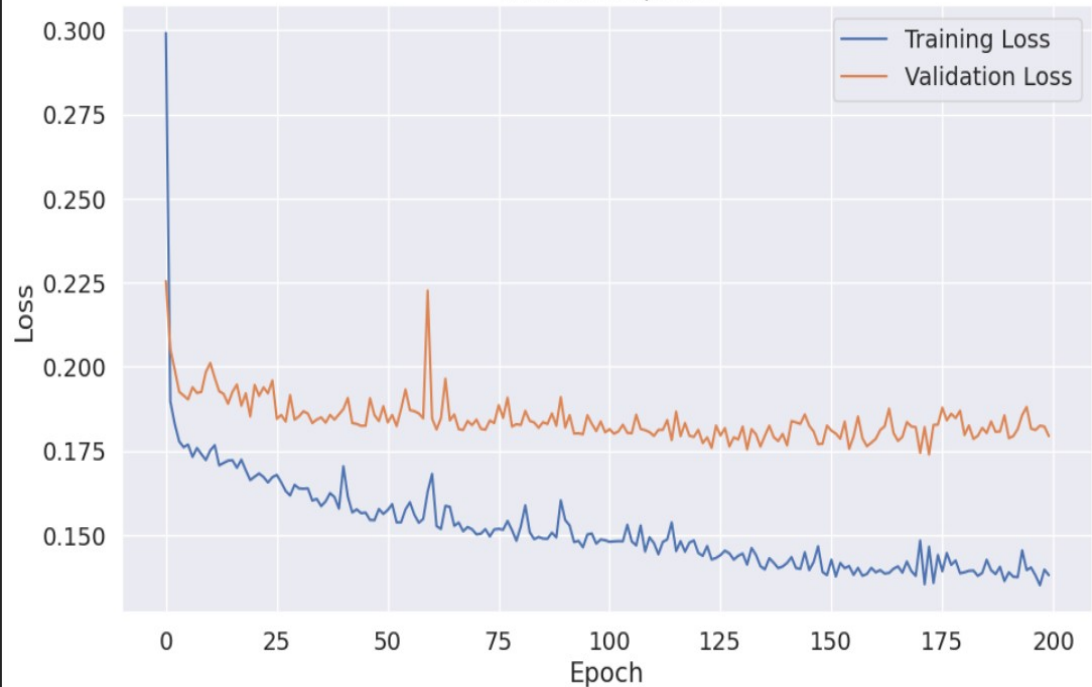
Metric	Value
Accuracy	0.9454
Precision (Class 0)	0.95
Precision (Class 1)	0.76
Recall (Class 0)	0.99
Recall (Class 1)	0.46
F1-Score (Class 0)	0.97
F1-Score (Class 1)	0.58
Macro Average Precision	0.86
Macro Average Recall	0.72
Macro Average F1-Score	0.77

Neural Networks Model

Confusion Matrix (Neural Network - Train Set)



Loss Over Epochs

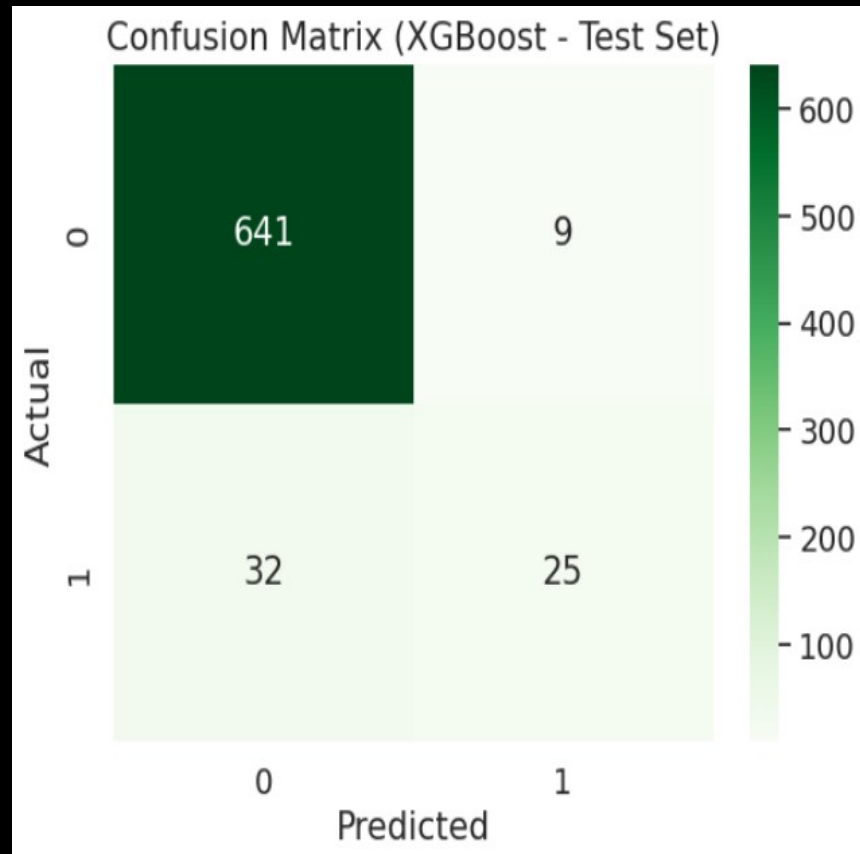


Comparison Of All the Models

Metric	Logistic Regression	Decision Tree	XGBoost	Neural Network
Accuracy	0.9229	0.9441	0.9454	0.94
Precision (class 1)	0.69	0.73	0.76	0.83
Recall (class 1)	0.07	0.48	0.46	0.38
F1-score (class 1)	0.12	0.58	0.58	0.52
Macro Avg Precision	0.81	0.84	0.86	0.89
Macro Avg Recall	0.53	0.73	0.72	0.69
Macro Avg F1-Score	0.54	0.77	0.77	0.75

Preferred Model: XGBoost

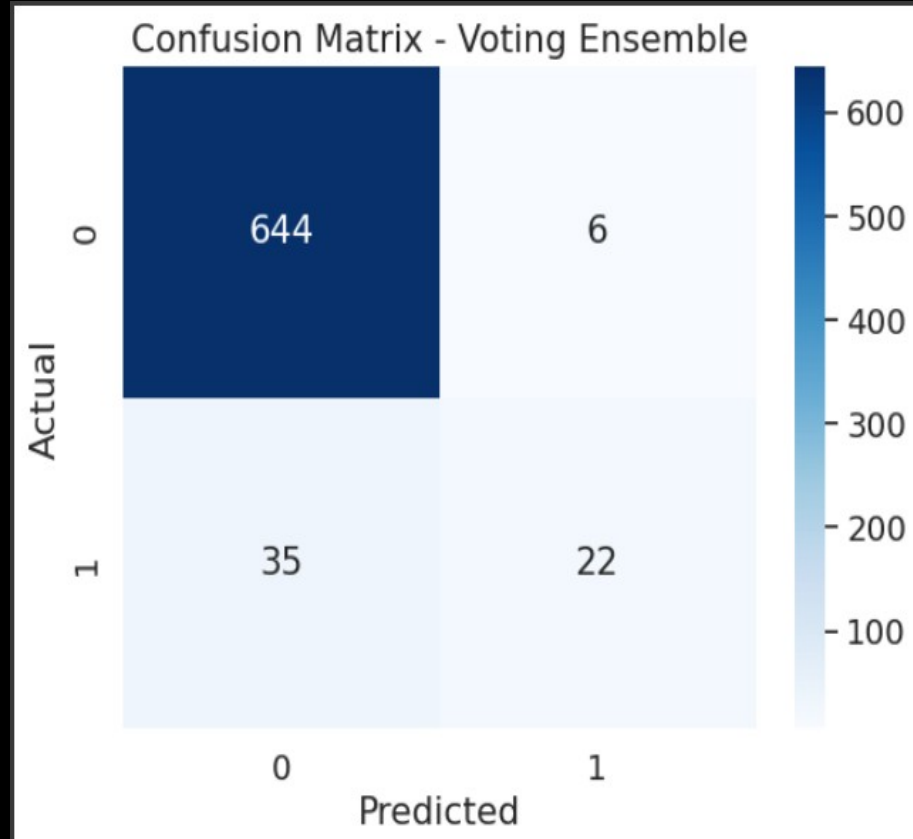
Class	Precision	Recall	F1-Score	Support
0	0.95	0.99	0.97	650
1	0.74	0.44	0.55	57
Accuracy			0.94	707
Macro Avg	0.84	0.71	0.76	707
Weighted Avg	0.93	0.94	0.94	707



Ensemble Model Evaluation

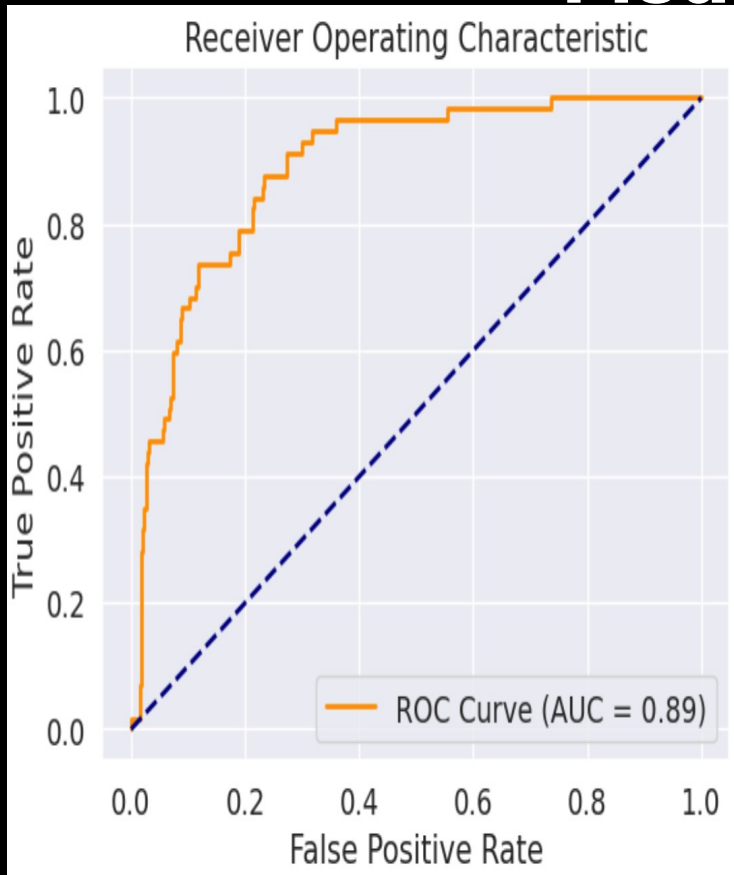
The Ensemble Model combines the predictions of multiple individual machine learning models to achieve more robust and accurate predictions than any single model. By integrating the strengths of different algorithms, the Ensemble Model aims to outperform the individual models and provide a more reliable and generalizable solution.

Class	Precision	Recall	F1-Score	Support
0	0.95	0.99	0.97	650
1	0.79	0.39	0.52	57
Accuracy			0.94	707
Macro Avg	0.87	0.69	0.74	707
Weighted Avg	0.94	0.94	0.93	707

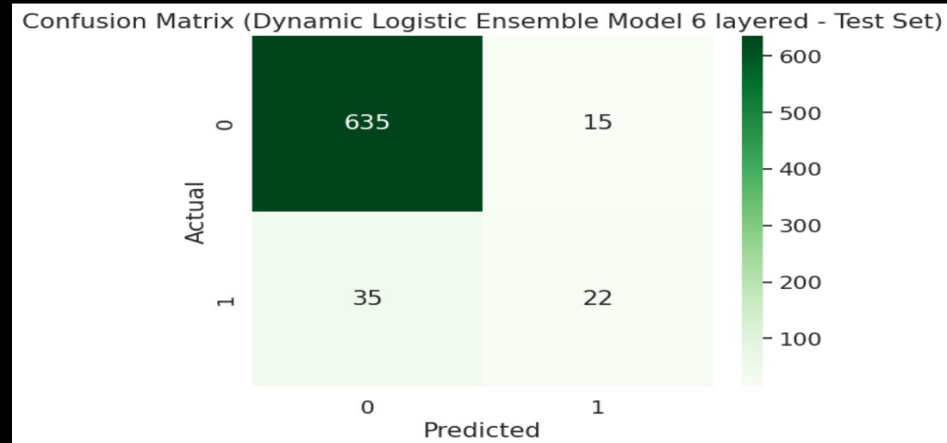


Dynamic Logistic Ensemble

Model - 3 Layers



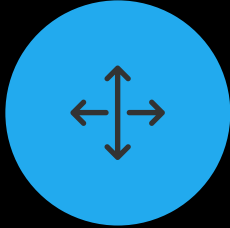
Class	Precision	Recall	F1-Score	Support
0	0.95	0.98	0.96	650
1	0.59	0.39	0.47	57
Accuracy			0.93	707
Macro Avg	0.77	0.68	0.72	707
Weighted Avg	0.92	0.93	0.92	707



Comparison of XG Boost, Ensemble Model, and Dynamic Logistic Ensemble Model

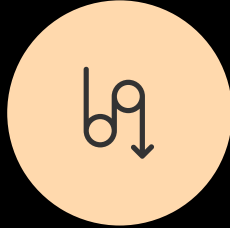
Metric	XGBoost	Ensemble Model	Dynamic Logistic Ensemble
Accuracy	0.9420	0.9420	0.9293
Precision (class 1)	0.74	0.79	0.59
Recall (class 1)	0.44	0.39	0.39
F1-Score (class 1)	0.55	0.52	0.47
Macro Avg Precision	0.84	0.87	0.77
Macro Avg Recall	0.71	0.69	0.68
Macro Avg F1-Score	0.76	0.74	0.72

Challenges & Solutions



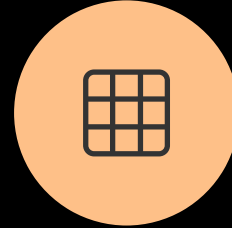
Class Imbalance

Minority class of hedging adopters made up only a small portion of the dataset, posing a challenge for model performance.



Hyperparameter Tuning

Optimized hyperparameters for tree-based models like XGBoost to balance accuracy and recall for the minority class.



Ensemble Methods

Implemented a Voting Classifier ensemble to combine multiple models and enhance the robustness of predictions.

By addressing class imbalance, tuning hyperparameters, and utilizing ensemble techniques, the model was able to overcome key challenges and deliver accurate predictions of hedging adoption in Russian banks.

Conclusions & Key Insights



Significant Financial Indicators Identified

Key financial metrics like log-transformed Securities, EBT, and Loans, as well as bank Reorganization status, were found to significantly impact hedging adoption.



Accurate Hedging Prediction Possible

The XGBoost model achieved 95% accuracy in predicting whether a bank will adopt swap-hedging, validating the ability to reliably forecast this behavior.



Ensemble Method Competitive Performance

The ensemble voting classifier, combining multiple machine learning models, showed competitive performance compared to the individual XGBoost model.

The analysis identified key financial indicators that drive hedging adoption, demonstrated accurate predictive capabilities, and validated the use of ensemble methods to handle the class imbalance challenge. These insights can help banks make more informed decisions about adopting swap-hedging strategies.

Thank you

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